Each solution below restates the question briefly, then works through the answer. Notation matches the lecture 1 notes: A^C for the complement, Ω for the outcome space, and $P(A \mid B) = P(A \cap B)/P(B)$.

Part 1: Set Theory

1) Rolling a die

$\Omega = \{1, 2, 3, 4, 5, 6\}$, with A = even and B = 4 or greater.

(a) $A = \{2, 4, 6\}$ and $B = \{4, 5, 6\}$.

(b) $A \cup B$ is everything in either set: $\{2, 4, 5, 6\}$.

(c) First $A^C = \{1, 3, 5\}$. Intersecting with $B = \{4, 5, 6\}$ keeps only the shared outcome: $A^C \cap B = \{5\}$.

(d) $A \cup B = \{2, 4, 5, 6\}$, so its complement is everything left over: $(A \cup B)^C = \{1, 3\}$.

2) Drawing a card

H = heart, F = face card (J, Q, K).

(a) A heart **and** a face card: $H \cap F$.

(b) A heart **but not** a face card: $H \cap F^C$.

(c) **Neither** a heart nor a face card: $(H \cup F)^C$. (Equivalently $H^C \cap F^C$ — the outcomes outside *both* sets.)

(d) On a two-circle Venn diagram this is the entire region **outside both circles**, i.e. everything in Ω that is not in H and not in F.

3) Shading a Venn diagram

(a) $A \cap B^C$: shade the part of circle A that does **not** overlap B (the left crescent).

(b) $(A \cup B)^C$: shade everything **outside both** circles.

(c) "Exactly one of A or B" is the two non-overlapping crescents — in A or in B, but not the overlap. In set notation:

$$\left(A \cap B^C\right) \cup \left(A^C \cap B\right).$$

4) Venn diagram with probabilities

$P(A) = 0.5$, $P(B) = 0.7$, $P(A \cap B) = 0.4$.

(a) Two overlapping circles; the overlap has probability 0.4.

(b) By inclusion–exclusion,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.5 + 0.7 - 0.4 = 0.8.$$

You cannot just add $P(A) + P(B) = 1.2$ because that counts the overlap $A \cap B$ **twice** (once inside A, once inside B), so you subtract one copy of it.

(c) Complement rule: $P\left(A^C\right) = 1 - P(A) = 1 - 0.5 = 0.5$.

(d) "A but not B" is A with the overlap removed:

$$P\left(A \cap B^C\right) = P(A) - P(A \cap B) = 0.5 - 0.4 = 0.1.$$

5) Mutually exclusive or not?

Two events are mutually exclusive when $A \cap B = \emptyset$ (they share no outcomes).

(a) Heart and spade: **mutually exclusive** — a single card can't be both.

(b) Heart and king: **not** mutually exclusive — the King of hearts is in both.

(c) Even and odd: **mutually exclusive** — no number is both.

(d) Even and 4-or-greater: **not** mutually exclusive — 4 and 6 are in both.

Part 2: The Axioms of Probability

6) Inclusion–exclusion (coffee and tea)

Let A = drinks coffee, B = drinks tea.

(a) The problem gives $P(A) = 0.7$, $P(B) = 0.4$, and $P(A \cap B) = 0.2$. It does **not** give $P(A \cup B)$ directly — we have to compute it.

(b) $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.7 + 0.4 - 0.2 = 0.9$.

(c) "Neither" is the complement of "coffee or tea":

$$P\left((A \cup B)^C \right) = 1 - 0.9 = 0.1.$$

(d) "Coffee but not tea" is A with the overlap removed:

$$P\left(A \cap B^C\right) = P(A) - P(A \cap B) = 0.7 - 0.2 = 0.5.$$

7) Which axiom breaks?

(a) **Impossible.** If A and B were mutually exclusive, Axiom 3 would give $P(A \cup B) = 0.6 + 0.7 = 1.3$. But no probability can exceed $P(\Omega) = 1$ (Axiom 2), so this can't happen.

(b) **Impossible.** $P(A) = -0.2$ violates Axiom 1, which requires $P(A) \geq 0$.

(c) **Possible.** Here $P(A \cup B) = 0.6 + 0.7 - 0.4 = 0.9 \leq 1$, and every probability stays in $[0, 1]$. No axiom is violated.

Part 3: Conditional Probability

8) Commuter table

	Bike	Bus	Car	**Total**
Late	0.05	0.15	0.05	**0.25**
On time	0.20	0.25	0.30	**0.75**
Total	**0.25**	**0.40**	**0.35**	**1.00**

(a) Read the margins: $P(\text{Late}) = 0.25$ and $P(\text{Bus}) = 0.40$.

(b) The single cell where Late meets Bus: $P(\text{Late} \cap \text{Bus}) = 0.15$.

(c) $P(\text{Late} \mid \text{Bus}) = \frac{P(\text{Late} \cap \text{Bus})}{P(\text{Bus})} = \frac{0.15}{0.40} = 0.375$.

(d) $P(\text{Bike} \mid \text{Late}) = \frac{0.05}{0.25} = 0.20$ and $P(\text{Bike} \mid \text{On time}) = \frac{0.20}{0.75} \approx 0.267$. A commuter is **more likely to bike when on time** ($0.267 > 0.20$).

(e) **Not independent.** $P(\text{Late} \mid \text{Bus}) = 0.375$ but $P(\text{Late}) = 0.25$. Knowing someone took the bus raises their chance of being late, so the two events are related.

9) Comparing $P(A)$, $P(A \mid B)$, and $P\left(A \mid B^C\right)$

$P(A) = 0.3$, $P(B) = 0.5$, $P(A \cap B) = 0.2$.

(a) $P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{0.2}{0.5} = 0.4$. Since $0.4 > 0.3 = P(A)$, conditioning on B made A **more** likely.

(b) First $P\left(A \cap B^C\right) = P(A) - P(A \cap B) = 0.3 - 0.2 = 0.1$, and $P\left(B^C\right) = 1 - 0.5 = 0.5$. So

$$P\left(A \mid B^C\right) = \frac{0.1}{0.5} = 0.2.$$

(c) **Not independent**, because $P(A \mid B) = 0.4 \neq 0.3 = P(A)$. A and B are positively associated: when B happens, A becomes more likely ($0.3 \to 0.4$); when B doesn't happen, A becomes less likely ($0.3 \to 0.2$).

10) Rain and umbrella (multiplication rule)

(a) $P(\text{rain and umbrella}) = P(\text{rain})\, P(\text{umbrella} \mid \text{rain}) = 0.4 \times 0.8 = 0.32$.

(b) $P(\text{no rain and umbrella}) = P(\text{no rain})\, P(\text{umbrella} \mid \text{no rain}) = 0.6 \times 0.1 = 0.06$.

(c) "Rain" and "no rain" are mutually exclusive and cover every case, so add the two pieces:

$$P(\text{umbrella}) = 0.32 + 0.06 = 0.38.$$